QIS digit classification

Beta–Bernoulli BOCPD and adaptive integration

Problem formulation

Each QIS recording contains a sequence of binary photon-detection frames.

INPUT

$B_t(x, y) \in \{0, 1\}$

Pixel coordinates: 0 to 27
Time index: binary frame
Loaded cube: [H, W, T]

OBJECTIVE

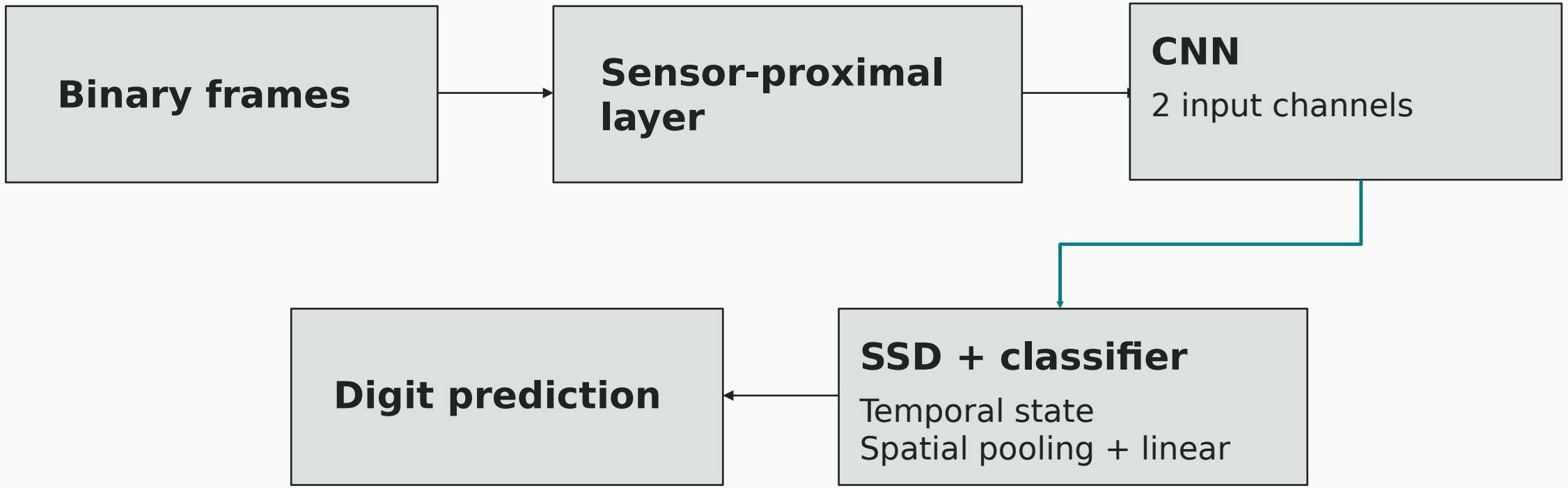
Classify the digit at each readout

Use a CNN and temporal SSD layer to follow the digit label through a photon-frame sequence.

The sensor-proximal layer supplies spatial readouts without a separate conventional reconstruction stage.

Pipeline overview

The integrator processes every binary frame. The CNN receives two maps at each saved readout.



Binary photon tensor

Each recording uses a dense sequence of binary frames.

Photon cube [B, H, W, T]

Dimension	Description	Implementation
H, W	Pixel location	28 × 28 pixels
T	Binary-frame time axis	Length of the loaded cube
B	Independent sequences	DataLoader batch size

Synthetic digit transitions

The evaluation morph creates a randomized wipe between two precomputed photon cubes.

Two recordings: digit A and digit B

Phase	Source selection	Target label	Timing
Before wipe	Mostly A	A	Before start
During wipe	A/B pixels	B	About 100 frames
After wipe	Mostly B	B	After the wipe

Binary photon detection model

Assume a locally constant detection probability at each pixel.

BINARY OBSERVATION

$$B_t \in \{0, 1\}$$

$$B_t \mid p \sim \text{Bernoulli}(p)$$

$$p = 1 - \exp(-N_t)$$

UNCERTAINTY ABOUT p

$$p \mid \alpha, \beta \sim \text{Beta}(\alpha, \beta)$$

α: detection evidence
β: non-detection evidence
Mean probability: α / (α + β)

What each forecaster stores

By default, eight forecasters track possible times of the latest probability change at each pixel.

Stored quantity	Description	Used for
Birth frame / age r_s	How long this regime may have lasted	Integration-time estimate
Beta parameters α_s, β_s	Uncertainty about detection probability	Predictive likelihood
Normalized weight ν_s	Relative support for this hypothesis	Weighted run-length estimate

Update at each binary frame

Both detections and non-detections update the forecasters at every pixel.

SCORE THE BINARY OBSERVATION

$$L_s(B_t) = \frac{\alpha_s B_t + \beta_s (1 - B_t)}{\alpha_s + \beta_s}$$

UPDATE HYPOTHESIS SUPPORT

Continuing weight: $(1 - \gamma)\ \nu_s\ L_s(B_t)$

New candidate: $\gamma\ \Sigma_s\ \nu_s\ L_s(B_t)$. Normalize the retained weights.

UPDATE THE BETA PARAMETERS

$$\alpha_s \leftarrow \alpha_s + B_t;\ \beta_s \leftarrow \beta_s + 1 - B_t$$

Run length and adaptive update weight

Compute an integration length from the forecaster state before each binary-frame update.

WEIGHTED AGE ESTIMATE

$$\hat{r} = \exp\left[\Sigma_s \nu_s \ln(1 + r_s) \right] - 1$$

Run lengths use frame units. A spatial minimum filter shares shorter integration lengths across nearby pixels.

$$w_t = 1 - \exp(-1 / \text{minpool}(\hat{r}_t))$$

Adaptive exponential integration

Each binary frame updates a smoothed signal and its adaptive update-weight map.

$$I_t = (1 - w_t)\, I_{t-1} + w_t\, B_t$$

Large w_t

Faster response and less temporal averaging

Small w_t

Longer memory and stronger temporal averaging

Second CNN channel

$$\tilde{w}_t = w_t \,/\, (\max_{x,y} w_t + \varepsilon)$$

Normalize the map separately for each sample and saved frame.

Processing one binary frame

The current frame uses the previous forecaster state to set its integration weight.

State	Update uses	Effect
Integrated signal I_t	Current B_t and weight w_t	Current spatial readout
Forecaster bank	Current B_t and Beta likelihood	Next frame's weight

Temporal subsampling

D sets the readout cadence. Every raw frame still updates the sensor-proximal state.

Illustrative sequence	Value
Raw frames T	1000 frames
Physical frame period	Not specified
Subsampling factor D	20 frames
Saved readouts	50 readouts

BOCPD + integrator

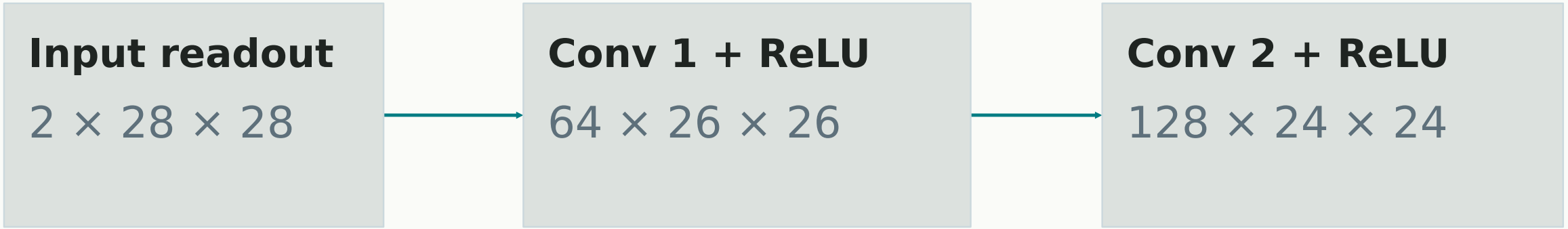
Process every binary frame

CNN + SSD

Process saved readouts

Spatial feature extraction

Each saved readout supplies the integrated signal and the normalized update-weight map.



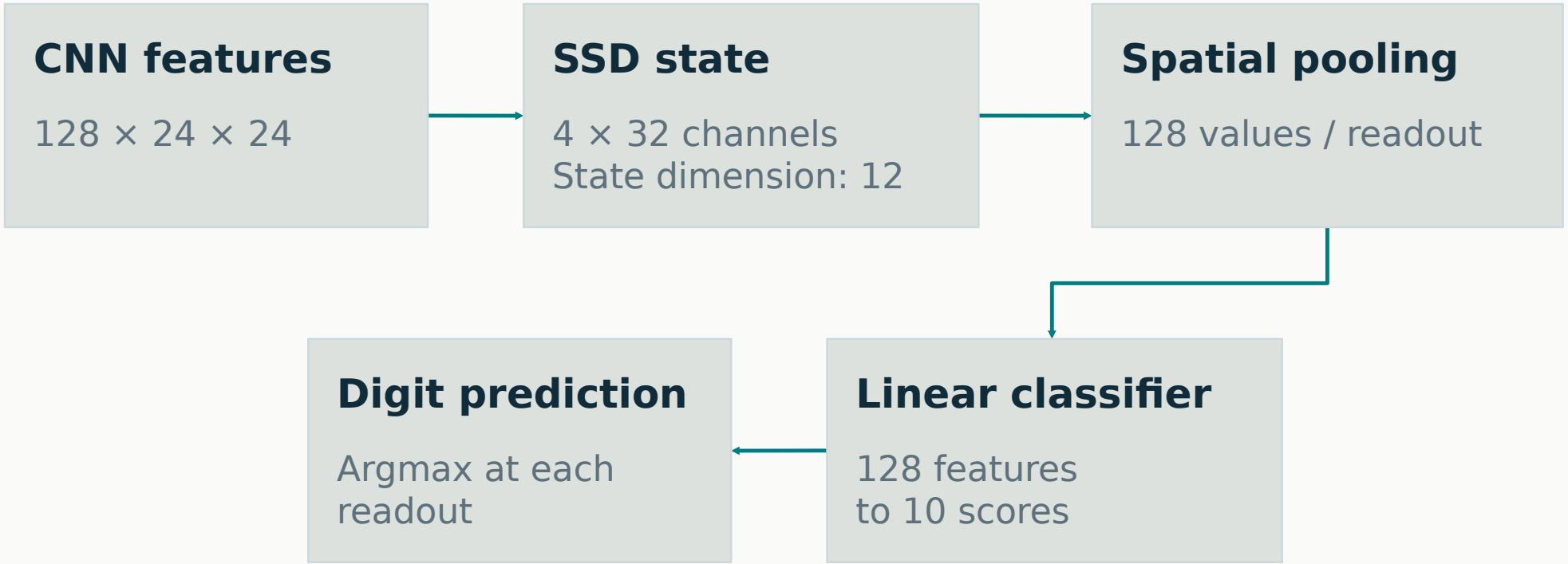
BOTH CONVOLUTIONS

3 × 3 kernel
Stride 1
No padding

Channel 1: integrated signal. Channel 2: normalized update weight.

Temporal features and classification

The SSD processes the feature sequence. The classifier outputs scores at every readout.



Training and evaluation

Both modes start each independent photon cube with fresh model state.

Operation	Training	Evaluation
Input delivery	Whole morphed cube	Whole morphed cube
SSD execution	Sequence forward pass	One recurrent step per readout
Target labels	A / ignore 100 frames / B	A then B at wipe start
Objective	Cross-entropy, smoothing 0.1	Per-readout digit accuracy

Training and evaluation currently use different transition-label conventions.